

Erdős Problem #700

EP-700

L1 - TRACTABLE

PARTIALLY SOLVED

Number Theory

ORIGINAL PROBLEM STATEMENT

Let $f(n) = \min_{1 < k \leq n/2} \gcd\left(n, \binom{n}{k}\right)$. {UL}

{LI}Characterise those composite n such that $f(n) = n/P(n)$, where $P(n)$ is the largest prime dividing n .{/LI}

{LI}Are there infinitely many composite n such that $f(n) > n^{1/2}$?{/LI}

{LI}Is it true that, for every composite n , $f(n) \ll_A \frac{n}{(\log n)^A}$ for every $A > 0$?{/LI}

{/UL}

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Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>